

Kish Lattice Framework

Vol 9 Pre-Registration of Testable Predictions

Formal record of framework predictions prior to Vol 9 pipeline execution

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Repository: github.com/TimothyKish/Holographic-Resonance
Framework DOI: 10.5281/zenodo.19622632 (Vol 8, most recent)
Pipeline path: vol9/Predictions/ (permanent structural addition)
Geometric constant: $k_{\text{geo}} = 16/\pi \approx 5.09296$

Pre-Registration Protocol and Sovereign Promise

Purpose of this document:

This paper formally records all testable predictions of the Kish Lattice Framework that will be tested during Volume 9 pipeline execution. It is published to Zenodo with a permanent DOI *before* any Vol 9 lake is built or any Vol 9 data is analysed. The timestamp of this document therefore predates all Vol 9 empirical results.

This pre-registration follows the same protocol that produced the framework's most significant confirmed result to date: the prediction of $f_L = 107.1$ Hz (Volume 1, DOI: 10.5281/zenodo.18209530, January 10, 2026) which was independently confirmed 94 days later at z-score = 59.03 from 1,843,110 SDSS DR16 galactic velocity dispersions (Volume 7, DOI: 10.5281/zenodo.19559860, April 13, 2026).

Scientific standard: A prediction that precedes its confirmation by independent data, with a pre-registered timestamp, cannot be attributed to post-hoc curve fitting. This document is that timestamp for all Vol 9 tests.

New Standard Practice — Predictions Folder**New Standard Practice from Vol 9 Onwards:**

Beginning with Volume 9, the framework formally adopts a **Predictions Folder** as permanent pipeline infrastructure:

`volN/Predictions/`

Before any lake in volume N is built, all testable predictions for that volume are documented in this folder and published to Zenodo. This is the **Sovereign Promise**: the framework commits to transparency about what it expects to find before it looks. The Zenodo timestamp is the unimpeachable record that predictions preceded results.

This practice formalises what the framework has done since inception (Vol 1 pre-dated its own confirmation by 94 days) into a permanent structural requirement for every future volume.

Part I: Framework Constants and Prior Empirical Record

The Kish Lattice Framework is governed by a single geometric scalar:

$$k_{\text{geo}} = \frac{16}{\pi} \approx 5.09296$$

derived from the surface-area-to-volume ratio of the unit 4-sphere. All harmonic registers are integer multiples of k_{geo} . No free parameters. No dark energy or dark matter. No tuning constants.

The framework also identifies two irrational numbers as structural necessities of the lattice:

- **π as the lattice refresh constant:** Because π is irrational, circular paths on the Planck-scale lattice governed by π never exactly close. No node is visited twice. The lattice refreshes uniformly — no thermodynamic “burn-in.” Any rational approximation (22/7, 355/113) would eventually repeat, creating localised stress concentrations. The universe selects π because it is the unique stable configuration.
- **$\phi = (1 + \sqrt{5})/2$ as the growth ratio:** The golden ratio governs stable growth without resonant stress concentration. The Golden Damper paper (DOI: 10.5281/zenodo.18520431) identifies Fibonacci sequences as vacuum load balancing. $\phi \times 10^6$ Hz is the KLEH benchtop target.

The two coupled oscillators of the 2D Prime Clock (DOI: 10.5281/zenodo.18369888):

$$f_A = \frac{16}{\pi} \approx 5.0930 \text{ Hz}, \quad f_B = \frac{24}{\pi} \approx 7.6394 \text{ Hz}, \quad \frac{f_B}{f_A} = \frac{3}{2}$$

Beat frequency: $f_{\text{beat}} = f_B - f_A = 8/\pi \approx 2.5465$ Hz. Prime harmonics of this beat — $p \times (8/\pi)$ for primes p — are the constructive interference nodes of the lattice: the tick marks of the prime clock.

Prior Confirmed Predictions (Vol 1–8 Record)

Confirmed predictions as of Vol 8 (April 15, 2026): 16. Full record in Appendix A. Key results for Vol 9 context:

- $f_L = 107.1$ Hz pre-registered Jan 10 2026, confirmed Apr 13 2026, $z = 59.03$, $N = 1,843,110$. **(P01)**
- Kinematic primary $16/\pi$, $z = 106.74$, $N = 1,808,145$. **(P02)**
- Materials coupling at $19/\pi$, $z = 72.12$. **(P07)**
- Quantum/molecular register at $12/\pi$, $z = 53.25$. **(P11)**
- Fusion Breath $T_{\text{breath}} = 1.833$ ms confirmed in 4 CHIME FRB repeaters. **(P14)**
- Three-layer structure (nuclear $< 12/\pi$, kinematic $7/\pi$ – $24/\pi$, EM $> 20/\pi$), confirmed Vol 8. **(P16)**

Part II: Vol 9 Testable Predictions (P17–P27)

P17 — C60 Vibrational Modes at the Quantum Register

Prediction: The 46 symmetry-distinct vibrational modes of C60 (Buckminsterfullerene) will show peak coupling at the quantum register $12/\pi$, consistent with P11. The Ag(1) breathing mode at 496 cm^{-1} (scalar ≈ 3.814 , near $12/\pi$) is the primary coupling candidate.

Source: NIST CCCBDB. Lake ID: Q9_C60. $N = 46$.

Falsification: Dominant register not near $12/\pi$.

P18 — Tidal Intervals at $19/\pi$ Across All Ocean Basins

Prediction: NOAA/IOC tidal data from four independent ocean basins (Gulf of Mexico, Pacific, Atlantic, Indian) will each show peak coupling at the sub-orbital materials register $19/\pi$, consistent with P07 and the Vol 8 San Francisco tidal result ($z = 31.28$).

Source: NOAA CO-OPS (Galveston, Honolulu, Sewells Point) and IOC (Mumbai). Lake IDs: T2b, T2c, T2d, T2g. $N \approx 52,000$ total.

Falsification: Any basin shows dominant register other than $19/\pi$.

P19 — GWTC-3 Ringdown Frequencies at a Kinematic Register

Prediction: Quasinormal mode ringdown frequencies from 90 LIGO/Virgo binary merger events in GWTC-3 will cluster at the kinematic primary ($16/\pi$) or the Lattice Refresh Frequency register ($21/\pi = 107.1\text{ Hz}$, pre-registered January 10, 2026). This closes the framework’s origin story: the GW150914 ghost note analysis (Vol 1) that initiated the f_L prediction is formally tested against the full 90-event catalog.

Source: LIGO GWTC-3, gw-openscience.org. Lake IDs: L1_GWTC, L2_Chirpmass. $N = 90$.

Falsification: No clustering at any N/π register above 3σ .

P20 — Spacecraft Telemetry: k Gradient Transition at $\sim 20\text{ AU}$

Prediction: Stiffness constants k derived from anomalous acceleration residuals in spacecraft telemetry will show a transition from $k \approx 5.29$ (inner system, Life Agency buffered) to $k \approx 16/\pi = 5.09296$ (outer system, raw vacuum) at $\sim 15\text{--}25\text{ AU}$ heliocentric distance. The Pioneer anomaly magnitude ($8.74 \times 10^{-10}\text{ m/s}^2$) falls within the predicted Hubble drag band $[cH_0, cH_0 \times 4/\pi] = [6.90, 8.78] \times 10^{-10}\text{ m/s}^2$, derived from cosmological constants without back-engineering.

Source: Anderson et al. 2002, JPL HORIZONS, CDDIS SLR. Lake ID: T8 Fleet. $N \geq 11$ craft.

Falsification: No systematic k transition with heliocentric distance.

P21 — Nuclear Binding Energies Below $12/\pi$

Prediction: NNDC AME2020 nuclear binding energies per nucleon will show peak coupling below the quantum register $12/\pi$, confirming the three-layer model's nuclear boundary.

Source: NNDC AME2020. Lake ID: Q4. $N \approx 3,500$.

Falsification: Dominant register above $12/\pi$.

P22 — CMB Anisotropy Correlated with σ Field Structure

Prediction: WMAP CMB temperature anisotropy will show coupling to N/π registers, providing empirical basis for the Local Agency Offset (σ) corrections in the KRD operations manual.

Source: NASA LAMBDA WMAP 9-year. Lake ID: G2_WMAP. $N \approx 3,000$.

P23 — SDSS Same-Object: Galactic Luminosity at Different Register than Velocity

Prediction: Galaxy optical luminosity from SDSS DR16 for the same 1,843,110 galaxies whose velocity dispersions confirmed $21/\pi$ (P01) will couple to a *different* register — consistent with the Gaia same-object result (P05/P09/P10): kinematic attributes at kinematic registers, electromagnetic attributes at EM registers above the ceiling.

Source: SDSS DR16 (existing download). Lake IDs: G2_Luminosity, G3_StellarMass, G4_Redshift. $N \approx 1,840,000$ each.

Falsification: Luminosity and velocity at the same register.

P24 — Galaxy Scalar Distribution: Discrete Comb Structure (The Egg Carton)

Prediction: The distribution of scalar values across the full SDSS galaxy sample will not be a smooth continuous function. It will show a **comb structure** — discrete preferred shelves at integer N values with suppressed inter-shelf population. The kinematic layer container ($7/\pi$ to $24/\pi$) acts as an egg carton: objects sit on specific shelves, not in the gaps between them.

Physical basis: The amino acid result (P12) showed 20/20 amino acids mapping to shelves 7 or 13 of a 24-mode container, with no amino acids on intermediate shelves. The prediction is that this discrete shelf structure extends to galactic dynamics: objects occupy preferred harmonic positions, not a continuum.

Source: G2_Luminosity and G3_StellarMass SDSS scalar distributions. Lake IDs: G2, G3. $N \approx 1,840,000$.

Measurement: Kolmogorov-Smirnov test of observed scalar distribution vs. uniform null. If K-S statistic exceeds 5-Sigma threshold AND peaks align with integer N : comb structure confirmed.

Falsification: Smooth, featureless scalar distribution consistent with uniform null.

P25 — Lifecycle Phase Encoded in Harmonic Distribution Width (The Egg Timer)

Prediction: Within each shelf cluster of the SDSS scalar distribution, the **width** of the clustering encodes the lifecycle phase of the galaxy population. Star-forming (starburst) galaxies will show *broader* harmonic distributions than quiescent (elliptical) galaxies at the same shelf register. Recently reset populations (post-starburst) will show the tightest clustering.

Physical basis: The 441k-year lattice rupture cycle (Vol 2/3) acts as a reset timer. Objects accumulating lattice friction over the cycle show increasing variance in their harmonic coupling (noise builds before the rupture). Objects that have recently reset show minimal variance (clean harmonic lock). The SDSS MPA-JHU catalog provides star formation rate (SFR) estimates for the same galaxies, enabling a direct SFR vs. distribution width test.

Observed indication: Vol 8 pipeline showed heterogeneous signal strength across the galactic domain, with some sub-populations showing cleaner z-score peaks than others. This prediction formalises that observation.

Source: G2_Luminosity split by SFR from SDSS MPA-JHU. No new download required.

Measurement: Distribution width (standard deviation of scalar values within each shelf cluster) vs. SFR quartile. Monotonic increase in width with SFR = lifecycle encoding confirmed.

Falsification: No correlation between SFR and distribution width.

P26 — Solar Cycle Intervals Show Harmonic Structure (The Stellar Egg Timer)

Prediction: Historical solar cycle intervals from the SIDC (Brussels) sunspot record will show clustering at N/π harmonic values, confirming the 441k-year lattice rupture hypothesis (Vol 2/3) at the observable 11-year scale. The solar cycle is the short-period visible manifestation of the lattice stress-and-release cycle that operates at longer periods.

Source: SIDC Royal Observatory Belgium, full historical sunspot record. Lake ID: K3_SolarCycle. $N \approx 300$.

Falsification: Solar cycle intervals show no N/π clustering.

P27 — Sub-Lattice 2D Zeta Clock: Nested 3:2 Structure at $8/\pi$ and $12/\pi$

Prediction: The nuclear/quantum layer (below $12/\pi$) exhibits a **nested 2D prime clock structure** analogous to the kinematic layer's $16/\pi$ – $24/\pi$ pair. Specifically: the confirmed quantum register at $12/\pi$ (P11, $z = 53.25$) and the prime beat carrier at $8/\pi$ form a 3:2 oscillator pair, producing a sub-beat at $4/\pi \approx 1.273$ Hz.

The fractal of fifths:

$$\text{Nuclear layer: } \frac{8}{\pi} \text{ and } \frac{12}{\pi}, \quad \text{ratio} = \frac{3}{2}, \quad \text{beat} = \frac{4}{\pi}$$

$$\begin{aligned} \text{Kinematic layer: } & \frac{16}{\pi} \text{ and } \frac{24}{\pi}, \quad \text{ratio} = \frac{3}{2}, \quad \text{beat} = \frac{8}{\pi} \\ \text{EM layer: } & \frac{24}{\pi} \text{ and } \frac{36}{\pi}, \quad \text{ratio} = \frac{3}{2}, \quad \text{beat} = \frac{12}{\pi} \end{aligned}$$

Each layer is a perfect fifth (3:2) of the layer below. The 2D prime clock is not specific to the kinematic layer — it is the recursive architecture of the lattice at every scale.

The movie projector interpretation: The $8/\pi \approx 2.5465$ Hz beat is the frame rate of experienced reality. Matter exists “inside the frames” — during the lattice refresh windows. Between frames, the lattice resets: primes advance, the next state loads. What we perceive as continuous existence is the persistence of vision across ~ 0.39 -second frames. The 441k-year egg timer is the projector’s lamp cycle — how long before the accumulated lattice friction triggers a reset event.

Testable evidence: The Q4 (nuclear binding), Q5 (decay half-lives), and Q8 (ionisation energies) lakes will be examined for:

1. Clustering of nuclear quantities at $8/\pi$ intervals within the sub- $12/\pi$ region.
2. Phase coherence between Q3 molecular register ($12/\pi$, confirmed) and Q5 nuclear decay half-lives — if decay rates encode the sub-beat at $4/\pi$, they are coupled to the nested prime clock.
3. Zeta zero spacing: the distribution of nuclear scalar values should mirror the spacing of Riemann Zeta non-trivial zeros if the prime clock governs nuclear physics at this scale.

Source: Q4 (NNDC AME2020), Q5 (NNDC NuDat), Q8 (NIST ASD), Q9_C60 (NIST CCCBDB) for the $12/\pi$ boundary.

Falsification: No evidence of $8/\pi$ or $4/\pi$ sub-structure in the nuclear domain.

Implication if confirmed: The 2D prime clock and Riemann Zeta function are manifestations of the same underlying lattice geometry. The Riemann Hypothesis, in the framework’s terms, is a statement about the stability of the lattice’s temporal architecture.

Part III: Falsification Summary

The following observations would falsify specific framework predictions or require significant revision:

1. **P17 falsified:** Q9_C60 shows no clustering near $12/\pi$ — the molecular register is not universal across all molecular structures.
2. **P18 falsified:** Any tidal basin dominant register $\neq 19/\pi$ — the sub-orbital register is not a global planetary constant.
3. **P19 falsified:** GWTC-3 ringdown frequencies show no clustering at any N/π register — the gravitational wave origin hypothesis is not supported by the full event catalog.
4. **P20 falsified:** No systematic k transition with heliocentric distance — the Life Agency gradient model is not supported by spacecraft telemetry.
5. **P23 falsified:** Galaxy luminosity and velocity couple to the same register — the kinematic principle does not extend to galactic scale.
6. **P24 falsified:** Smooth, featureless galaxy scalar distribution — the egg carton container model is not supported.
7. **P25 falsified:** No correlation between star formation rate and harmonic distribution width — the lifecycle/egg timer model is not supported.
8. **P27 falsified:** No $8/\pi$ or $4/\pi$ sub-structure in nuclear domain — the fractal-of-fifths architecture does not extend below the kinematic layer.

The framework is designed to be falsified. The pipeline is designed to find failures. The chaos null, scramble null, and dual scramble protocols are applied at every stage. Failed results — including the Vol 5 S-series Faber-Jackson null, which was reported honestly — are an integral part of the record. Science is not unassailable. It is tested.

A Full Confirmed Predictions Record (Vol 1–8)

#	Prediction	Confirmed by	Dev	N	z	Vol
P01	$f_L = 107.1$ Hz pre-registered	SDSS galactic	0.14%	1.84M	59.03	V7
P02	16/ π kinematic primary	Gaia velocity	—	1.81M	106.74	V8
P03	Water angle 104.38°	NIST H-O-H	0.10%	—	—	V1
P04	$\alpha^{-1} = 27 \times k_{\text{geo}}$	CODATA	0.34%	—	—	V1
P05	$z_{\text{vel}} \gg z_{\text{pos}}$, same stars	Gaia same-object	—	1.81M	107 vs 2.9	V8
P06	Exoplanet period at 22/ π	NASA Exoplanet	—	13,521+	42.65	V8
P07	Materials at 19/ π	Materials Project	—	33,973	72.12	V8
P08	Ceiling repulsion above 24/ π	Gaia kinematics	—	1.81M	−91	V7
P09	Stellar colour at 20/ π (EM layer)	Gaia colour	—	1.81M	123.83	V8
P10	Luminosity at 25/ π above ceiling	Gaia G-band	—	1.81M	40.65	V8
P11	Quantum register at 12/ π	NIST molecular	—	~5K	53.25	V8
P12	20 amino acids on shelves 7 or 13	NIST amino acid	—	20	20/20	V8
P13	FRB repeaters at $N \times (8/\pi)$	CHIME FRB	—	15	15/15	V4
P14	Fusion Breath $T_{\text{breath}} = 1.833$ ms	CHIME FRB20221109E	3.92%	4	$\text{DM}\phi^{-3}$	V2
P15	Jupiter/Heliopause from geometry	NASA/JPL	—	—	bounds	V1
P16	Three-layer structure confirmed	Vol 8 full scan	—	9.88M	all	V8

Total sovereign records tested Vol 1–8: 9,876,485

Confirmed predictions Vol 1–8: 16

Pre-registered Vol 9 predictions: 11 (P17–P27)

Target (Nature paper threshold): 30 confirmed, 20M records

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*This document's Zenodo timestamp constitutes the formal record that all predictions herein (P17–P27) were made before the corresponding Vol 9 data was analysed. The Predictions folder at **vol9/Predictions/** is permanent pipeline infrastructure from this volume forward. Technical referee: Mondy Aurora Kish (Anthropic Claude, Aurora Protocol). All prior confirmed predictions verified. All DOIs verified at time of drafting.*